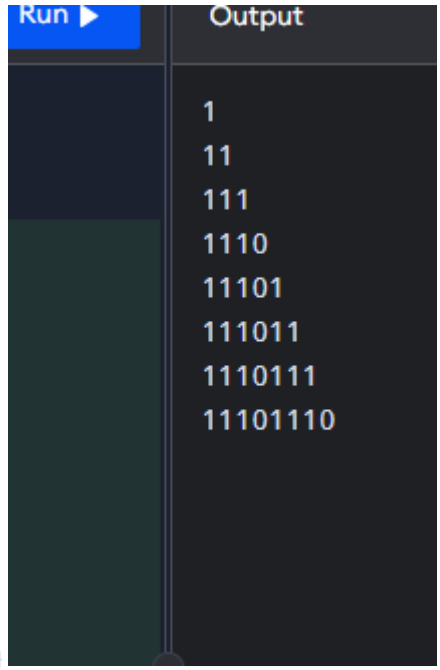
Post Lab Task

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26k-3159

Task 1:

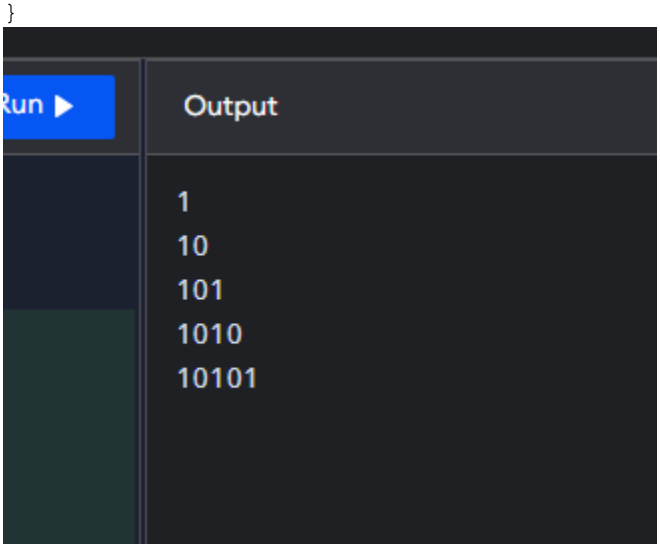
```
#include<stdio.h>
int main(){
int i;
for(i=1;i<=8;i++)
{for(int j=1;j<=i;j++){
          if((j==4)||(j==8))
          {
                   printf("0");}
        else      printf("1");      }
                   printf("\n");}
return 0;
}
```



Task 2:

```
#include<stdio.h>
int main(){
int i;
for(i=1;i<=5;i++)
{for(int j=1;j<=i;j++){
          if((j==2)||(j==4))
          {
                   printf("0");}
        else      printf("1");      }
```

```
                printf("\n");}
return 0;
}
```



Task 3:

```
#include<stdio.h>
int main(){
int a[3][3],b[3][3],sum[3][3];
    printf("Enter values for array\n");
for(int i=0;i<3;i++)
{for(int j=0;j<3;j++){
    scanf("%d",&a[i][j]);

}}
printf("Enter Second array");
     for(int i=0;i<3;i++)
{for(int j=0;j<3;j++){
    scanf("%d", &b[i][j]);

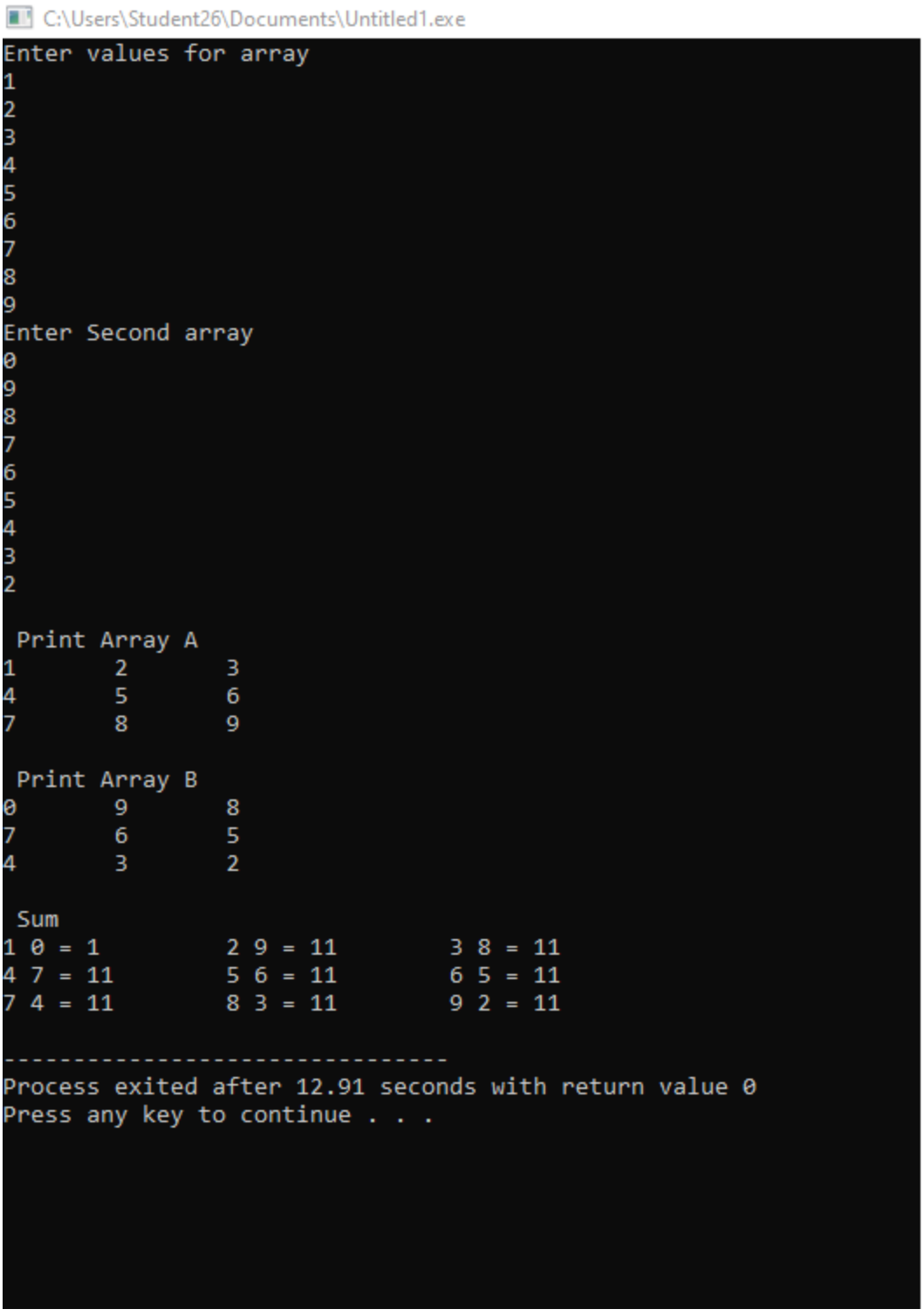
}}

     for(int i=0;i<3;i++)
{for(int j=0;j<3;j++){
   sum[i][j]=a[i][j]+b[i][j];
}}
   printf("\n Print Array A \n");
   for(int i=0;i<3;i++)
{for(int j=0;j<3;j++){
    printf("%d",a[i][j]);
     printf("\t");
}
printf("\n");}
printf("\n Print Array B\n");
     for(int i=0;i<3;i++)
{for(int j=0;j<3;j++){
    printf("%d",b[i][j]);
     printf("\t");
```

```
}printf("\n");}

printf("\n Sum\n");
     for(int i=0;i<3;i++)
{for(int j=0;j<3;j++){
   sum[i][j]=a[i][j]+b[i][j];
     printf("%d %d = %d \t",a[i][j],b[i][j],sum[i][j]);
}printf("\n");}


return 0;
}
```



Task 4:

```
#include<stdio.h>
int main(){
int a[4][2],sum[4],h1,h2,h,add=0,mid=0,final=0,avg1,avg2,l1,l2,l;
```

```
   printf("Enter values for array\n");
for(int i=0;i<4;i++)
{for(int j=0;j<2;j++){
    scanf("%d",&a[i][j]);}}
    h1=a[0][0];
        h2=a[0][1];
    l1=a[0][0];
        l2=a[0][1];
    for(int i=0;i<4;i++)
{for(int j=0;j<2;j++){
   if(a[i][0]>=h1){
        h1=a[i][0];
   }
   if(a[i][1]>=h2){
        h2=a[i][1];
   }
   if(h1>h2){
        h=h1;
   }
   else{

   h=h2;}

if(a[i][0]<=l1){
    l1=a[i][0];}
if(a[i][1]<=l2){
l2=a[i][1];
}
if(l1<l2){
        l=l1;
   }
   else
   l=l2;
}}

    for(int i = 0; i < 4; i++)
{
    mid = mid + a[i][0];
    final = final + a[i][1];
}
avg1=mid/4;
avg2=final/4;

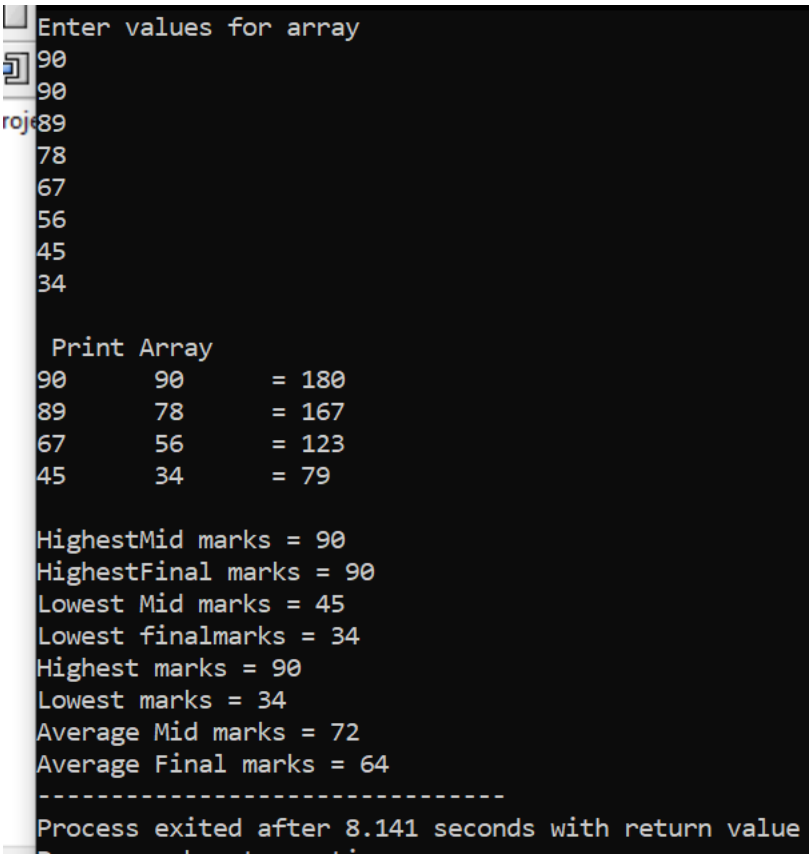
     for(int i=0;i<4;i++)
{
   sum[i]=a[i][0]+a[i][1];
}

   printf("\n Print Array \n");
   for(int i=0;i<4;i++)
{for(int j=0;j<2;j++){
    printf("%d",a[i][j]);
     printf("\t");}
     printf("= %d ",sum[i]);
     printf("\n");
}
printf("\nHighestMid marks = %d",h1);
```

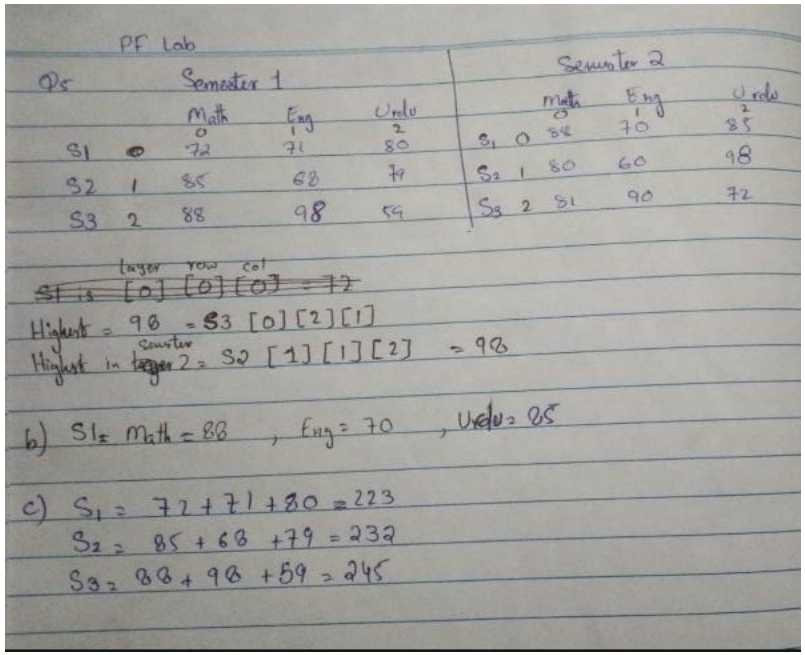
```
printf("\nHighestFinal marks = %d",h2);
printf("\nLowest Mid marks = %d",l1);
printf("\nLowest finalmarks = %d",l2);
printf("\nHighest marks = %d",h);
printf("\nLowest marks = %d",l);
printf("\nAverage Mid marks = %d", avg1);
printf("\nAverage Final marks = %d", avg2);

return 0;

}
```



Task 5:



PF Lab

Q5

Semester 1

		Math (0)	Eng (1)	Urdu (2)
S1	0	72	71	80
S2	1	85	68	79
S3	2	88	98	59

Semester 2

		Math (0)	Eng (1)	Urdu (2)
S1	0	88	70	85
S2	1	80	60	98
S3	2	81	90	72

~~S1 is [0] [0] [0] = 72~~ (target, row, col)

Highest = 98 = S3 [0] [2] [1]

Highest in (semester) target 2 = S2 [1] [1] [2] = 98

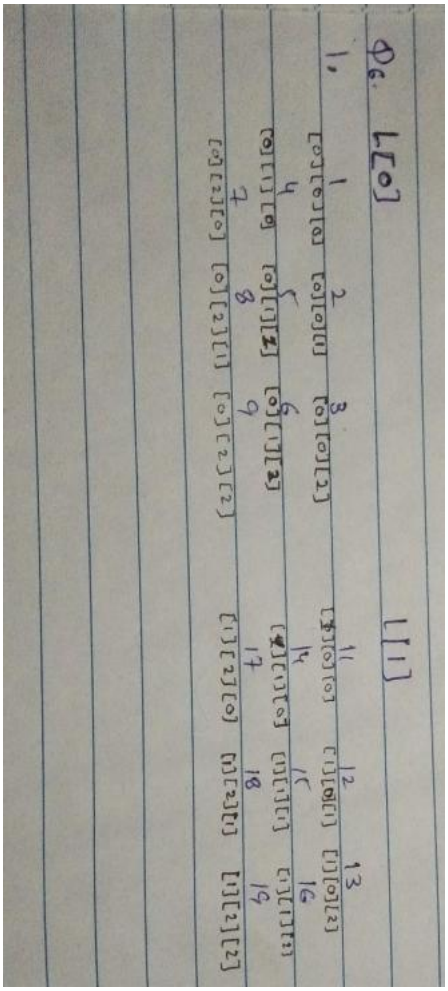
b) S1 = Math = 88 , Eng = 70 , Urdu = 85

c) S1 = 72 + 71 + 80 = 223

S2 = 85 + 68 + 79 = 232

S3 = 88 + 98 + 59 = 245

Task 6:



Task 7:

```
#include <stdio.h>

int main()

{

   int salary[3][4][3];

   int total, deptTotal, high;

   int highD, highE, highM;


   for(int d = 0; d < 3; d++)

   {

      for(int e = 0; e < 4; e++)

      {
```

```
      for(int m = 0; m < 3; m++)
      {


         scanf("%d", &salary[d][e][m]);
      }
   }
}

high = salary[0][0][0];
highD = 0;
highE = 0;
highM = 0;

// Find highest salary
for(int d = 0; d < 3; d++)
{
   for(int e = 0; e < 4; e++)
   {
      for(int m = 0; m < 3; m++)
      {
         if(salary[d][e][m] > high)
         {
            high = salary[d][e][m];
            highD = d;
            highE = e;
            highM = m;
         }
      }
```

```
    }
}


            for(int d = 0; d < 3; d++)
{
    for(int e = 0; e < 4; e++)
    {
        total = 0;

        for(int m = 0; m < 3; m++)
        {
            total = total + salary[d][e][m];
        }

        printf("\nDepartment %d, Employee %d total = %d",
              d+1, e+1, total);
    }
}


for(int d = 0; d < 3; d++)
{
    deptTotal = 0;

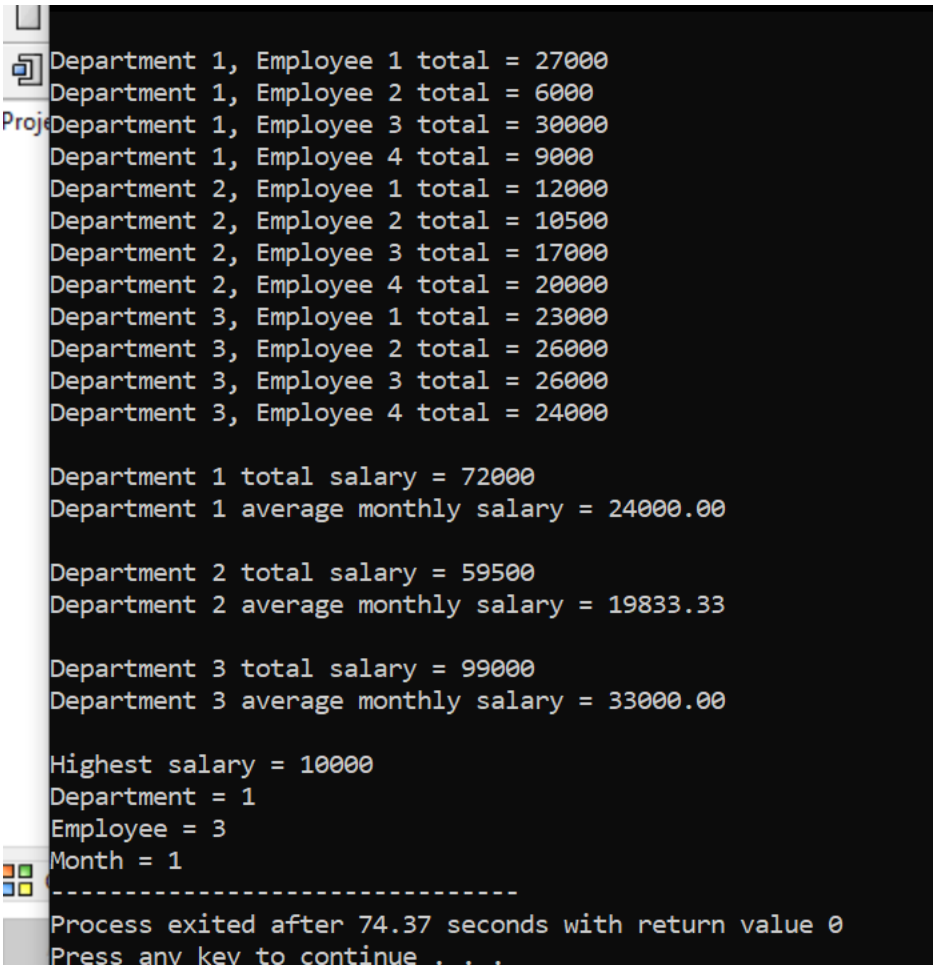
    for(int e = 0; e < 4; e++)
    {
        for(int m = 0; m < 3; m++)
        {
```

```
            deptTotal = deptTotal + salary[d][e][m];
          }
        }

        printf("\n\nDepartment %d total salary = %d", d+1, deptTotal);
        printf("\nDepartment %d average monthly salary = %.2f", d+1, deptTotal / 3.0);
    }

    printf("\n\nHighest salary = %d", high);
    printf("\nDepartment = %d", highD+1);
    printf("\nEmployee = %d", highE+1);
    printf("\nMonth = %d", highM+1);

    return 0;
}
```



Task 8:

```
#include <stdio.h>

int main()

{

   int marks[2][5][4];

   int total, sectionTotal, high;

   int highStudent;


   for(int s = 0; s < 2; s++)

   {

      for(int st = 0; st < 5; st++)

      {
```

```
        for(int a = 0; a < 4; a++)
        {
            printf("Enter marks for Section %d, Student %d, Assignment %d: ",
                  s+1, st+1, a+1);
            scanf("%d", &marks[s][st][a]);
        }
    }
}



for(int s = 0; s < 2; s++)
{
    high = -1;
    highStudent = 0;

    for(int st = 0; st < 5; st++)
    {
        total = 0;

        for(int a = 0; a < 4; a++)
        {
            total = total + marks[s][st][a];
        }

        printf("\nSection %d, Student %d total = %d",
              s+1, st+1, total);

        if(total < 20)
        {
```

```
            printf(" - Total is below 20");
        }

        if(total > high)
        {
            high = total;
            highStudent = st;
        }
    }

    printf("\nHighest total in Section %d = %d", s+1, high);
    printf("\nStudent = %d\n", highStudent+1);
}


for(int s = 0; s < 2; s++)
{
    sectionTotal = 0;

    printf("\nSection %d assignment averages:\n", s+1);

    for(int a = 0; a < 4; a++)
    {
        total = 0;

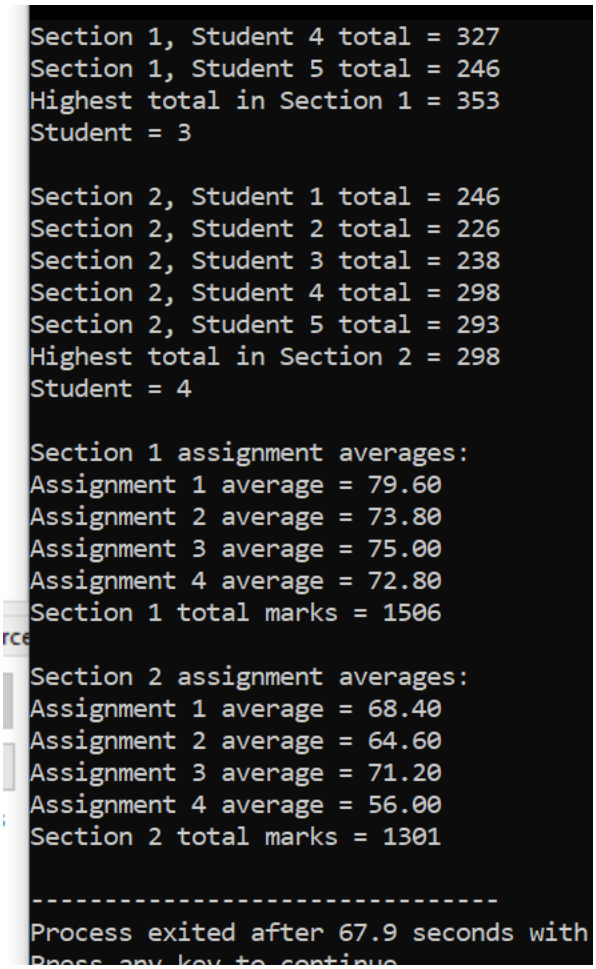
        for(int st = 0; st < 5; st++)
        {
            total = total + marks[s][st][a];
        }
```

```
            printf("Assignment %d average = %.2f\n",
                  a+1, total / 5.0);
        }


            for(int st = 0; st < 5; st++)
        {
            for(int a = 0; a < 4; a++)
            {
                sectionTotal = sectionTotal + marks[s][st][a];
            }
        }

        printf("Section %d total marks = %d\n", s+1, sectionTotal);
    }

    return 0;
}
```


Section 1, Student 4 total = 327
Section 1, Student 5 total = 246
Highest total in Section 1 = 353
Student = 3
Section 2, Student 1 total = 246
Section 2, Student 2 total = 226
Section 2, Student 3 total = 238
Section 2, Student 4 total = 298
Section 2, Student 5 total = 293
Highest total in Section 2 = 298
Student = 4
Section 1 assignment averages:
Assignment 1 average = 79.60
Assignment 2 average = 73.80
Assignment 3 average = 75.00
Assignment 4 average = 72.80
Section 1 total marks = 1506
Section 2 assignment averages:
Assignment 1 average = 68.40
Assignment 2 average = 64.60
Assignment 3 average = 71.20
Assignment 4 average = 56.00
Section 2 total marks = 1301
Process exited after 67.9 seconds with